OpenFlow/SDNとネットワーク仮想化

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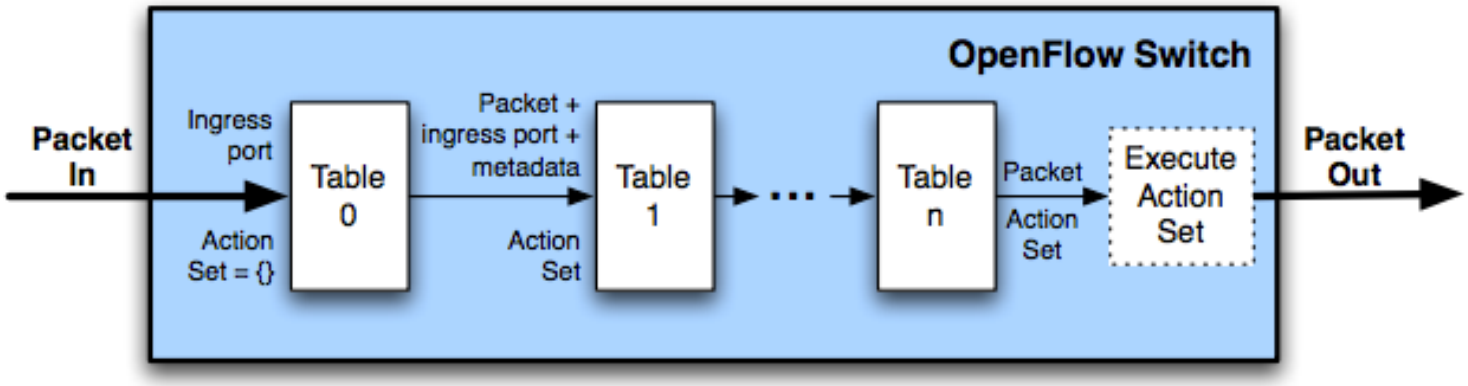
- ソフトウェアでコントロールする事？
- Control PlaneとData Planeの分離？
- 分散 vs 集中？
- OpenFlowを使う事?

- よりネットワークが「ソフトウェア」らしくなること
  - 垂直統合から水平統合へ
  - 開発・デバッグ手法
  - 配布モデル

- じゃ、OpenFlowって何よ？

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OpenFlow 1分間クッキング



Match Fields	Counters	Instructions

Table 1: Main components of a flow entry in a flow table.

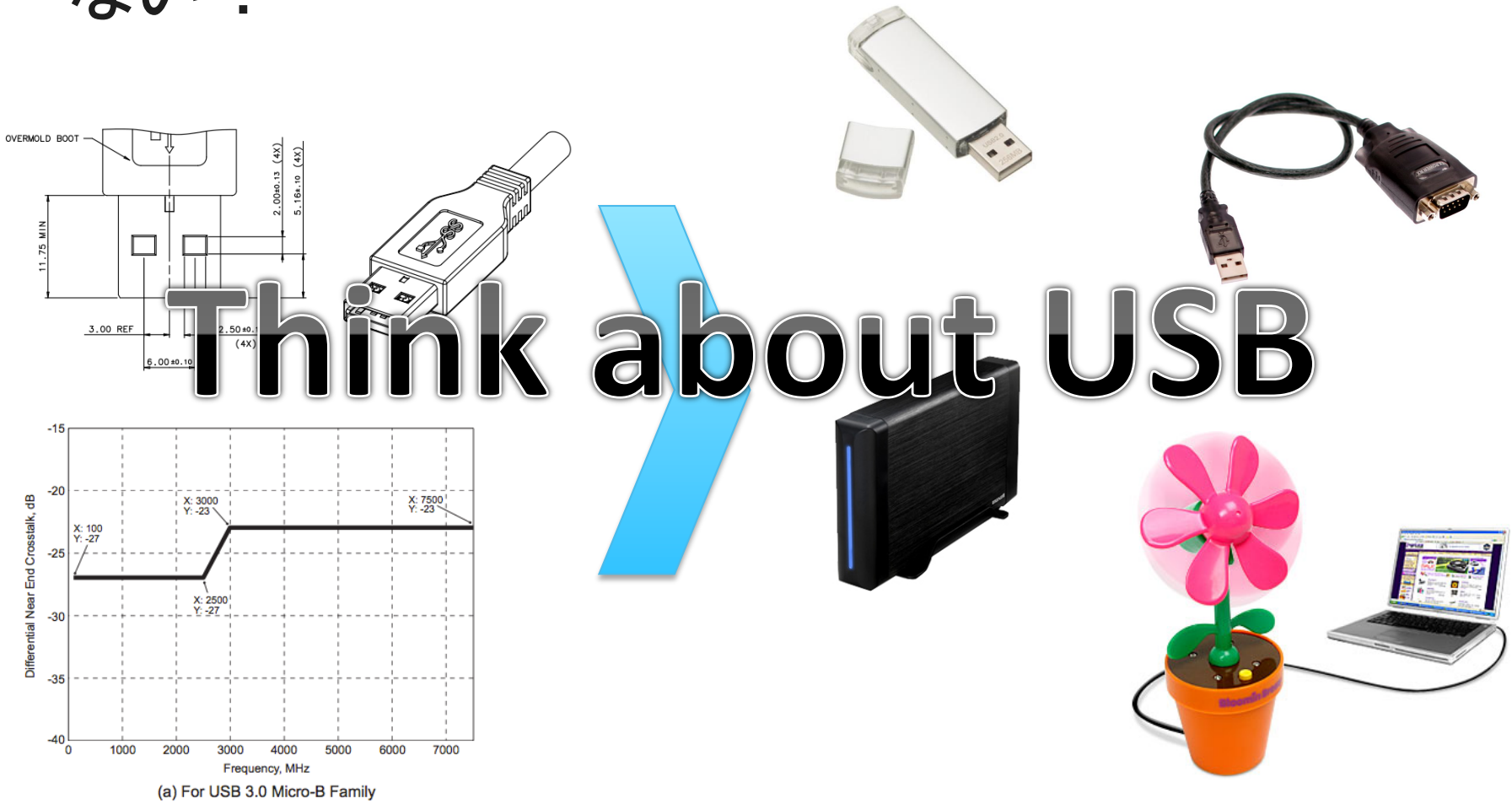
Ingress Port	Metadata	Ether src	Ether dst	Ether type	VLAN id	VLAN priority	MPLS label	MPLS traffic class	IPv4 src	IPv4 dst	IPv4 proto / ARP opcode	IPv4 ToS bits	TCP / UDP / SCTP src port	ICMP Type	TCP / UDP / SCTP dst port	ICMP Code

Table 3: Fields from packets used to match against flow entries.

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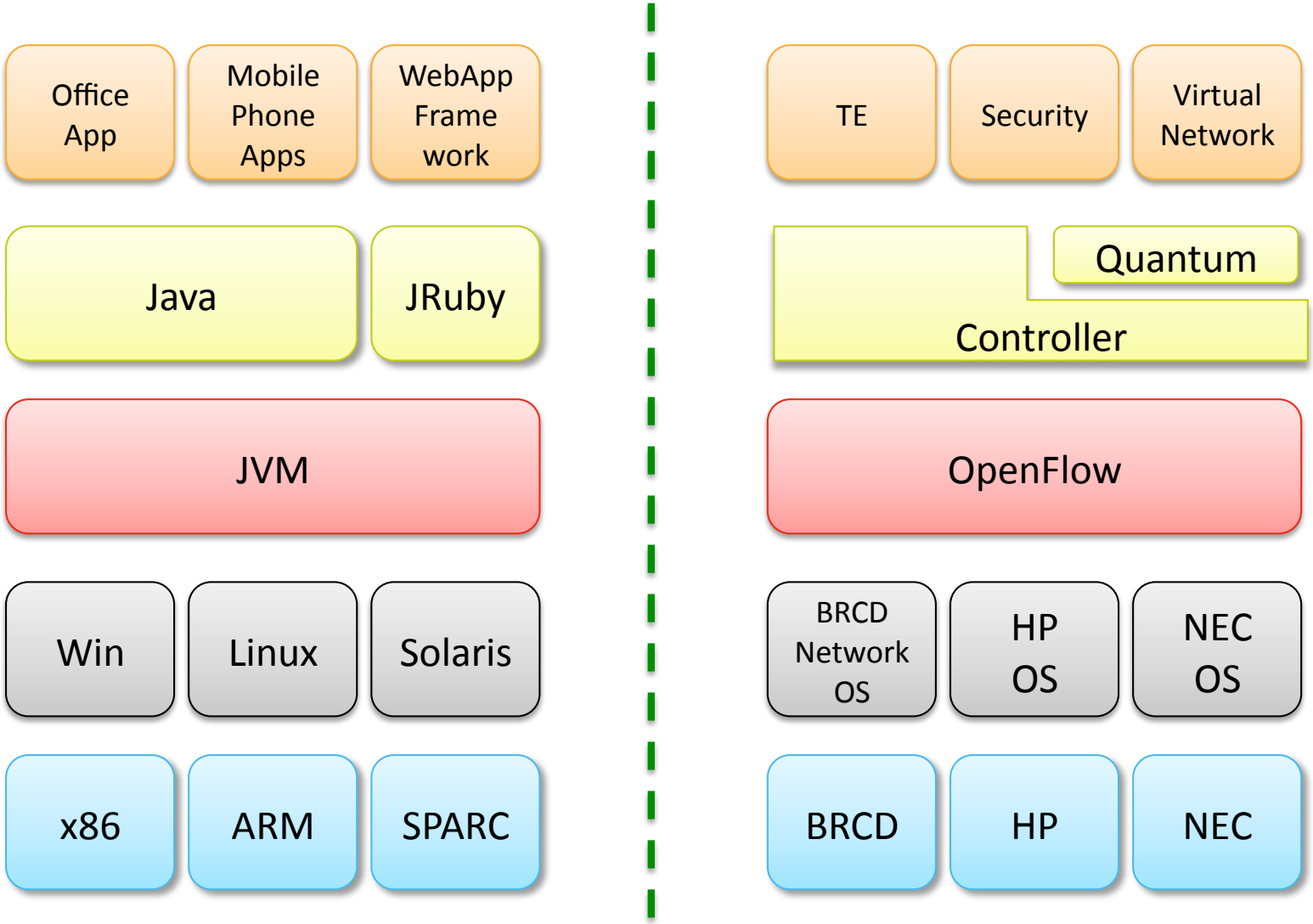
OpenFlowは・・・

- 便利な「ツール」であって「ソリューション」ではない！



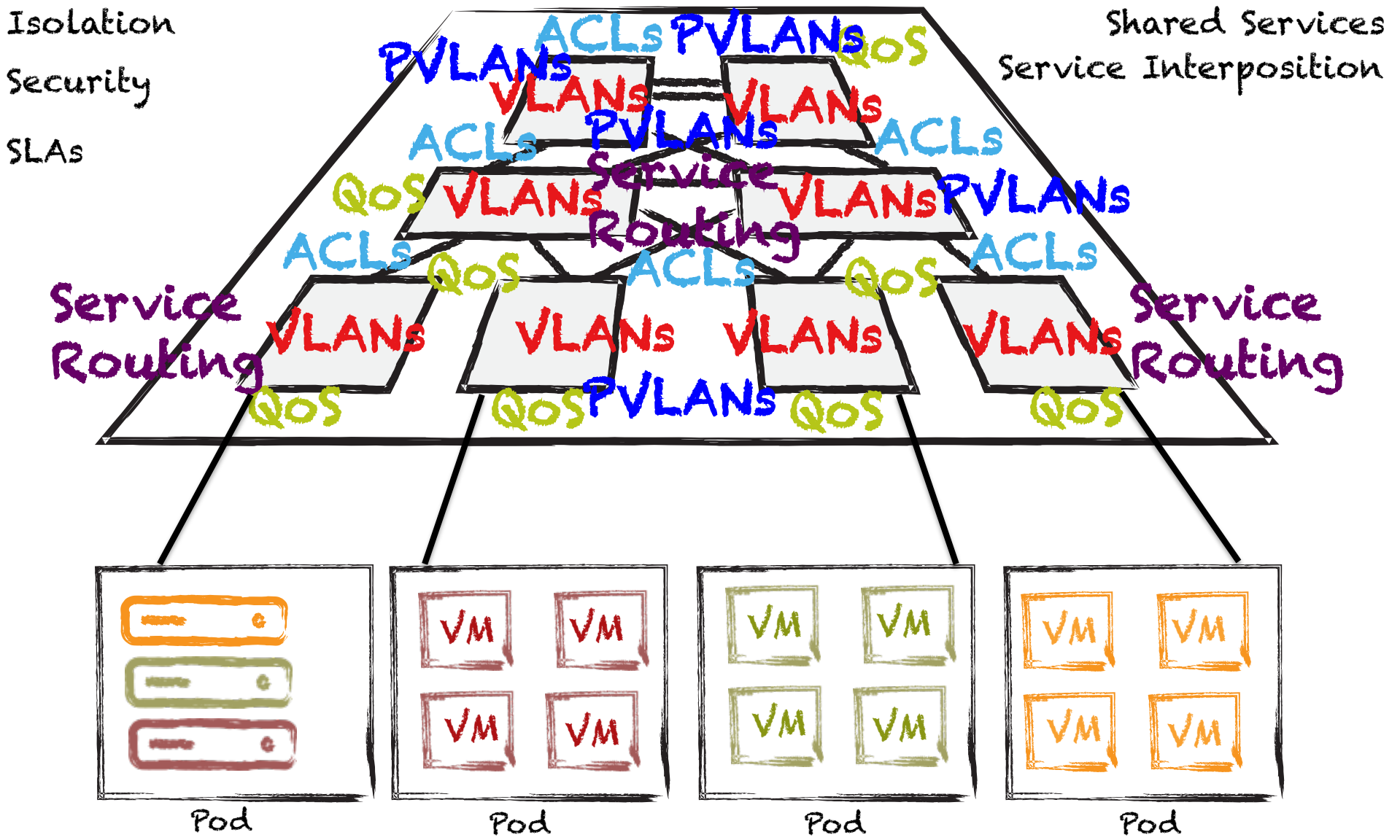
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Yet Another View of OpenFlow



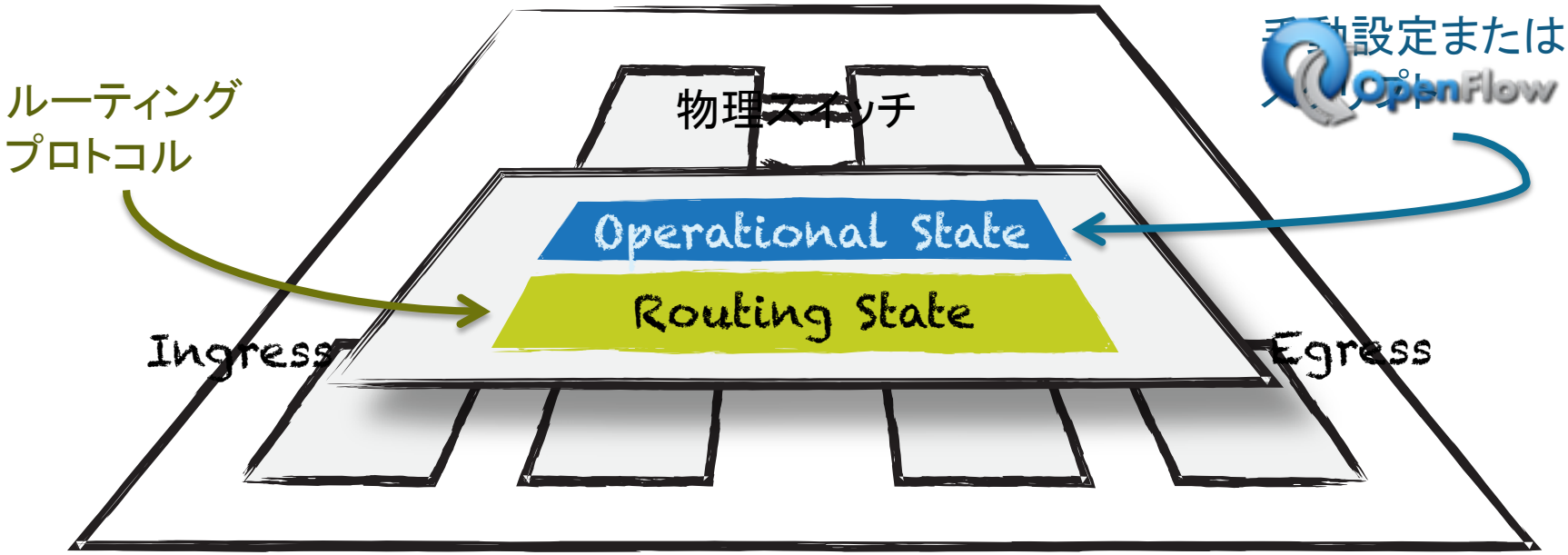
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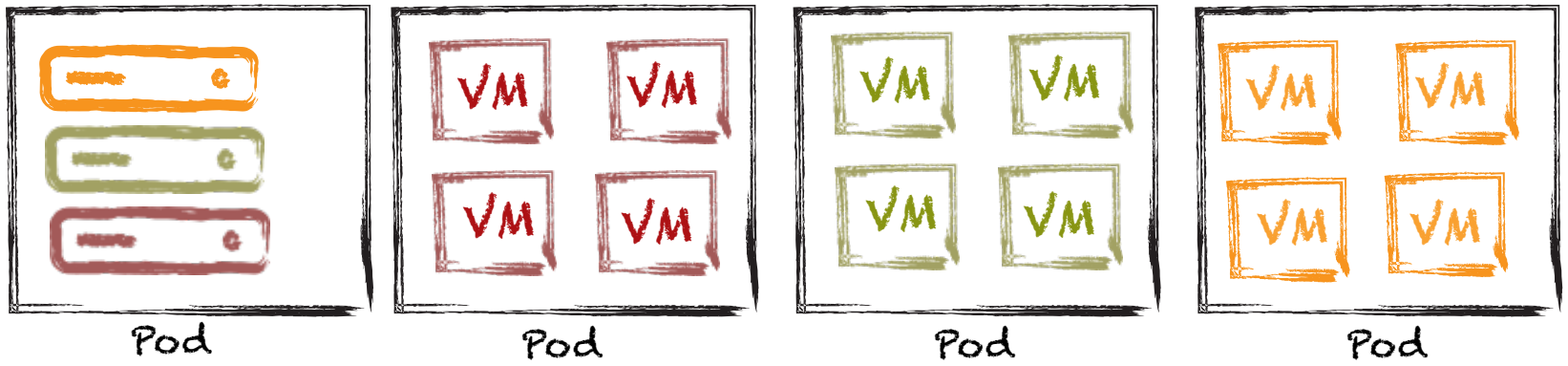
今日のネットワークは課題だらけ



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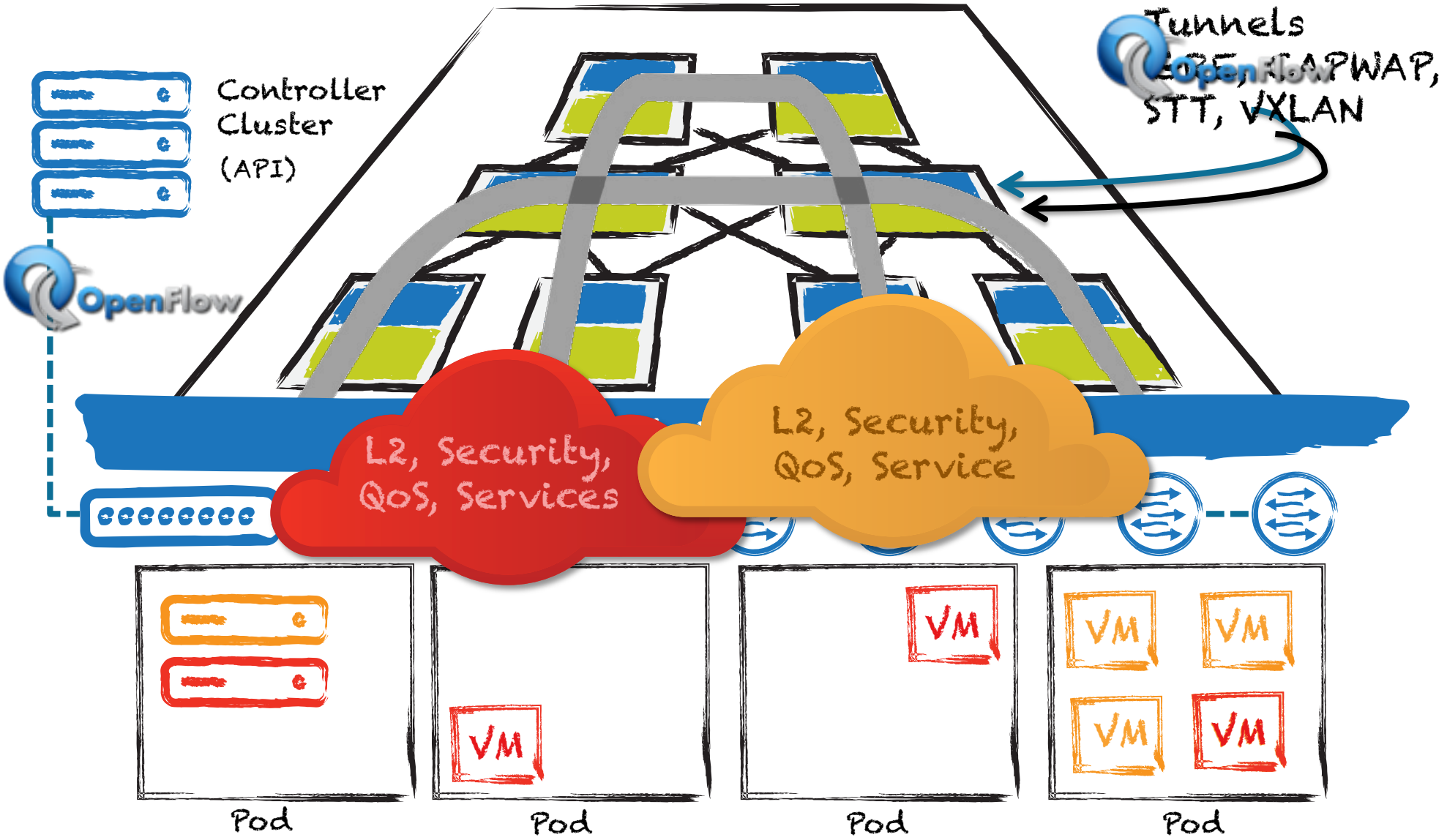
OpenFlowの生い立ち





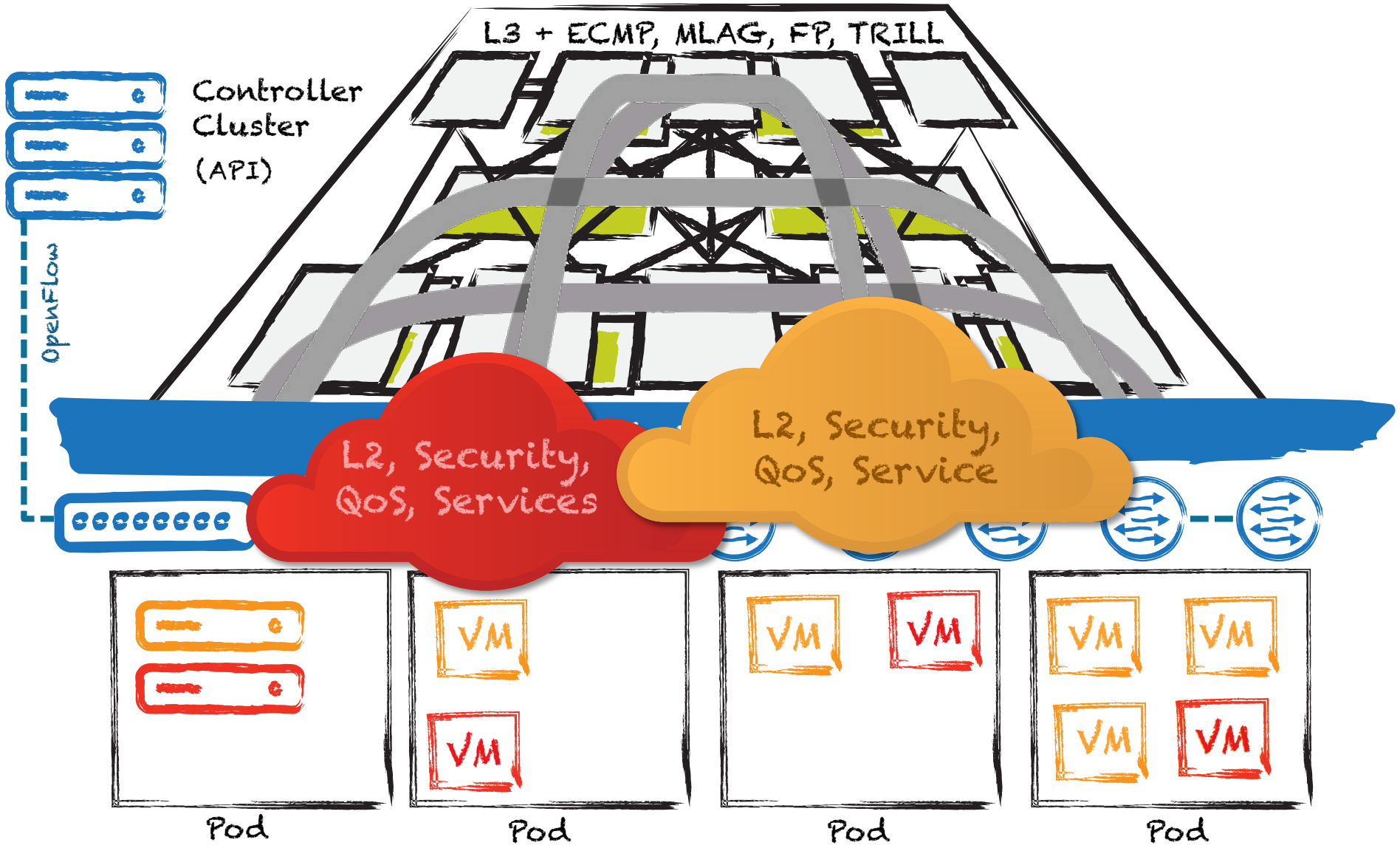
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仮想分散ネットワーク



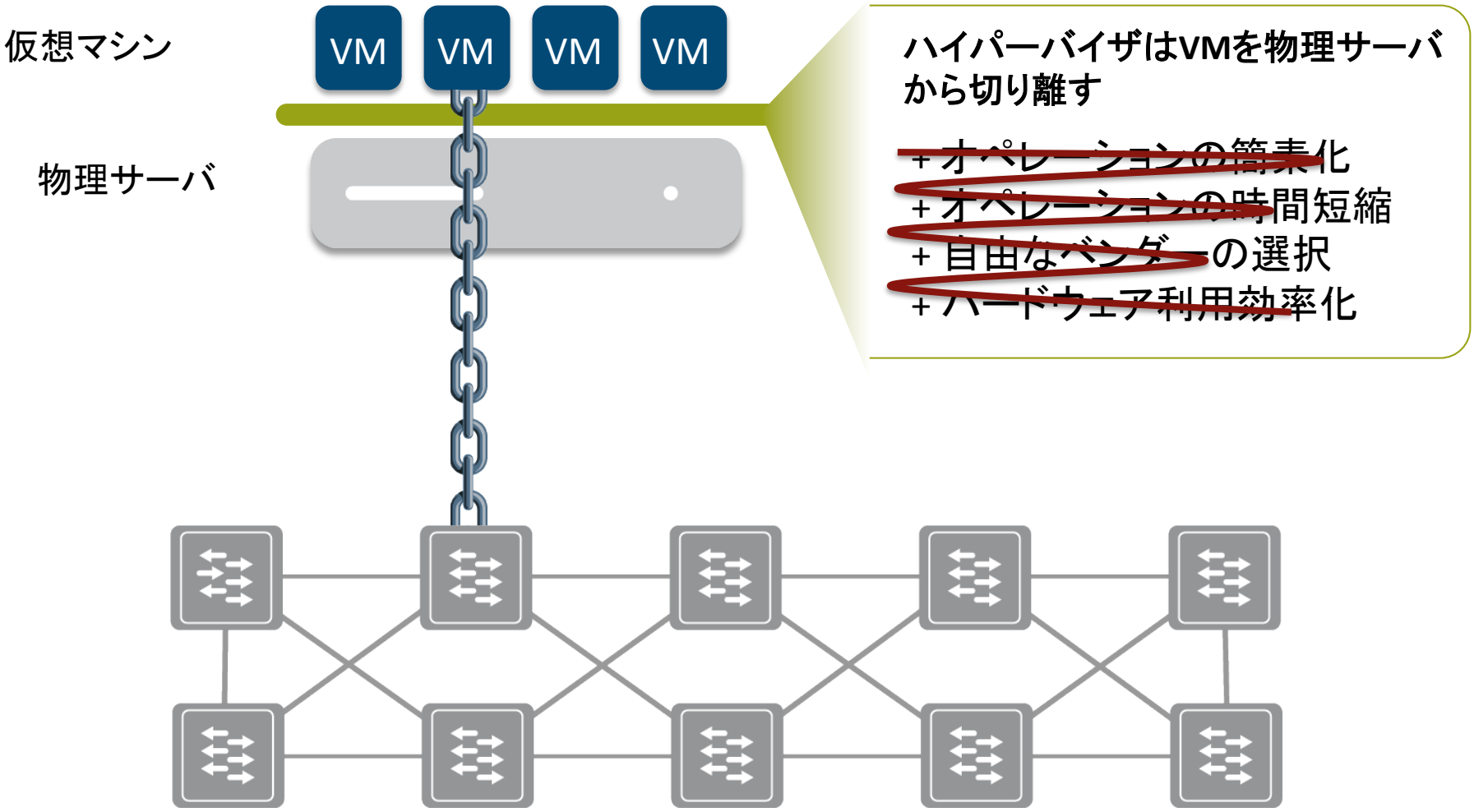
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ハードウェア独立性



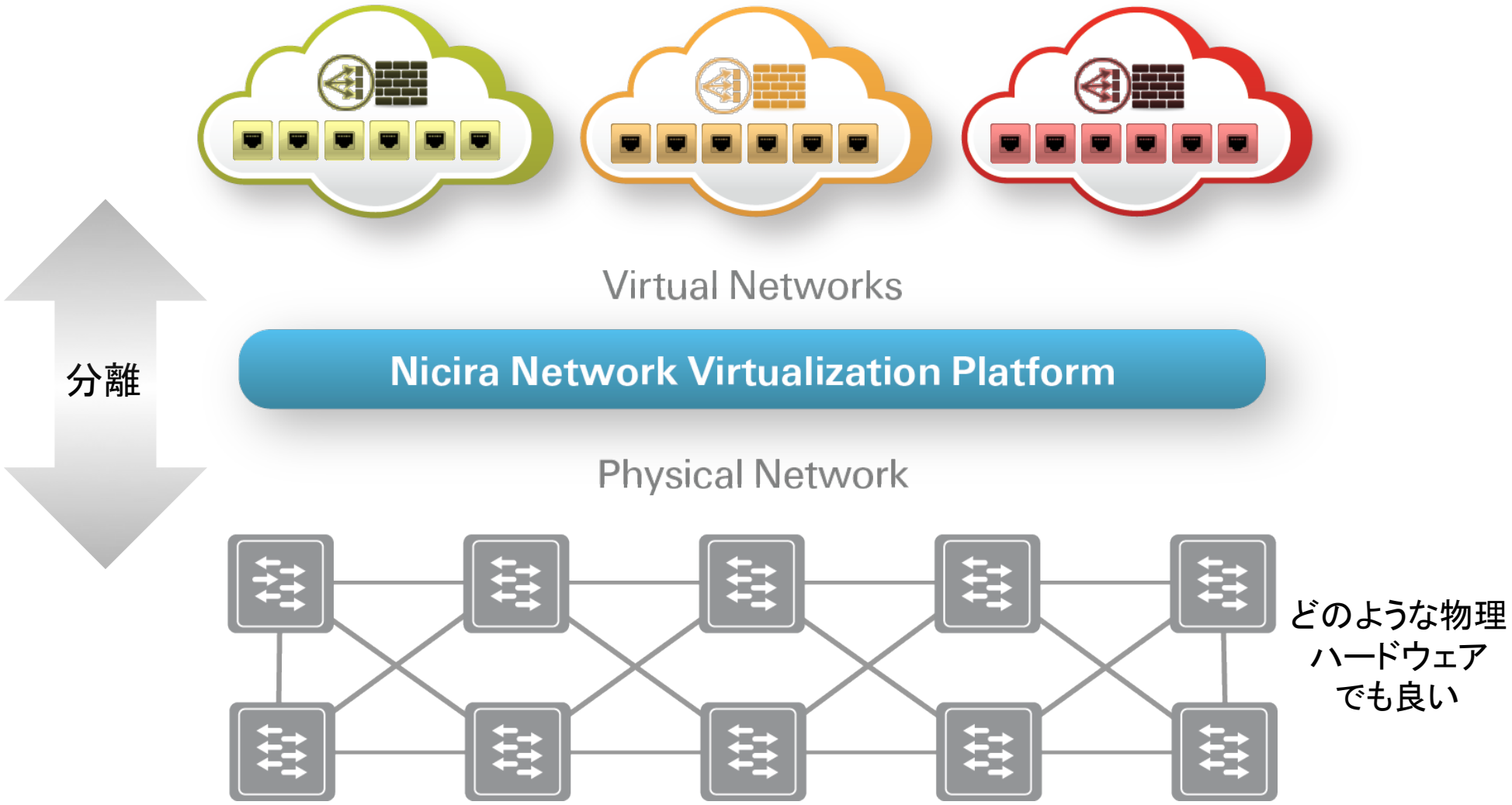
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ネットワークは今日のクラウドの障害



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NVPはネットワークを仮想化する



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まとめ

- SDNはネットワークの大きなパラダイムシフト
  - あらゆる面において「ソフトウェアっぽくなる」事
- OpenFlowはSDN/仮想ネットワークを実現するのに使える一つの便利なツール / インターフェース
- ネットワーク仮想化はSDNの（非常に有望な）1アプリケーション

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